

Additional Test Report — Core Tests and Non-Destructive Testing, Pier P5
Pier Shaft

Nandira River Bridge Project — Package NRB-4

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From:	Suvarna Materials Testing Laboratory
To:	Meridian Engineering Consultants
CC:	Sagara Constructions Pvt. Ltd.

Subject: Core Test and Ultrasonic Pulse Velocity (UPV) Results — Pier P5 Pier Shaft, Affected Zone (Test Set B)

Testing carried out 05-Apr-2022 to 08-Apr-2022, in response to NCR-NRB4-001 and Root Cause Analysis RCA-NRB4-003.

1. CORE TESTS: Three cores extracted from the affected zone of the Pier P5 pier shaft, length-to-diameter ratio 1.9 (correction factor 0.98 applied per the Specification's referenced core testing standard).

Core	Raw Strength (N/mm2)	Corrected Strength (N/mm2)
C-1	39.0	38.2
C-2	38.7	37.9
C-3	40.2	39.4
Average corrected core strength:		38.5 N/mm2

ASSESSMENT: All three corrected core results exceed 85 per cent of the specified characteristic strength (34.0 N/mm2), the criterion commonly applied for assessing in-place concrete strength from core results, and the average corrected strength approaches the specified characteristic strength itself.

2. ULTRASONIC PULSE VELOCITY (UPV) TESTING: UPV readings taken at 12 points across the affected zone and adjacent unaffected areas for comparison.

Zone	UPV Range (km/s)	Average (km/s)	Classification
Affected (Set B)	4.0 – 4.4	4.2	Good to Excellent
Adjacent (Set A/C)	4.1 – 4.5	4.3	Good to Excellent

ASSESSMENT: UPV readings across the affected zone are consistent with those in the unaffected zones and indicate dense, homogeneous concrete with no significant voiding, honeycombing, or discontinuity.

3. OVERALL CONCLUSION: Both the core test results and the UPV survey indicate the in-place concrete in the affected zone is sound and meets or closely approaches the specified characteristic strength, supporting the Root Cause Analysis conclusion that the low Test Set B cube results arose from deficiencies in the test

specimens rather than the in-place concrete.

Issued by: Suvarna Materials Testing Laboratory